Notes: Parrino, Sias, and Starks (JFE, 2003)

Voting with Their Feet: Institutional Ownership Changes around Forced CEO Turnover

Contents

1 Big picture

- **Research question.** Do institutional investors sell shares before forced CEO turnover, and does this "Wall Street Rule" behavior relate to board decisions about firing the CEO and choosing the successor?
- **Main idea.** Forced CEO turnover identifies a setting in which management is revealed, ex post, to have performed poorly enough that the board replaces the CEO. The paper uses this event to study whether institutional investors respond ex ante by reducing their positions.
- **Empirical object.** The main objects are changes in institutional ownership around CEO turnover: the change in the number of institutional investors and the change in the fraction of shares held by institutional investors. The paper studies both raw changes and abnormal changes adjusted for contemporaneous returns and marketwide institutional-ownership trends.
- **Main finding.** Institutional ownership falls substantially before forced CEO turnover. The largest selling occurs in the four quarters before the CEO is forced from office. The decline is not universal: many institutions buy while others sell, so the main outcome is a reshuffling of shareholder composition.
- **Who sells?** Selling is concentrated among institutions that appear more likely to be momentum traders, more concerned with holding prudent securities, or better informed. Bank trust departments and independent investment advisors are particularly important in the ownership shift.
- **Board response.** Measures of institutional selling are related to a higher likelihood of forced CEO turnover and to a higher likelihood that the replacement CEO comes from outside the firm.
- **Economic takeaway.** Institutional investors do not only govern through activism. Their trading decisions can also convey dissatisfaction, change the shareholder base, create price pressure, and potentially affect board actions.

2 Incremental contribution of the paper

- **From activism to exit.** The governance literature often studies shareholder activism. This paper studies exit: institutions may discipline or signal dissatisfaction by selling rather than by directly intervening.
- **Event-based setting.** Forced CEO turnover gives a clear window in which poor managerial performance is revealed. This allows the paper to examine investor behavior before the board acts.
- **Ownership composition, not just ownership level.** The paper shows that aggregate institutional ownership falls, but also that different types of institutions behave differently. The central phenomenon is a change in the identity of owners.
- **Multiple explanations for institutional selling.** The paper directly evaluates four explanations: momentum trading, prudence constraints, informed trading, and governance-cost concerns.

- **Connection to board decisions.** The paper links the ownership shift to forced turnover and outsider succession decisions, suggesting that trading by institutions may be relevant for corporate governance outcomes.
- **Granular institutional data.** CDA/Spectrum 13F data allow the authors to separate bank trust departments, insurance companies, investment companies, independent investment advisors, and other institutions.

3 Data and measurement

3.1 CEO turnover sample

The main sample covers CEO turnovers at large NYSE corporations from 1982 to 1993. The authors begin with 583 CEO turnovers that satisfy several requirements: both the departing and incoming CEOs appear in *Forbes* compensation surveys, the turnover is reported in the *Wall Street Journal*, Compustat data are available, the firm is listed on the NYSE, and the turnover is not directly related to a takeover.

- **Forced turnover.** A turnover is classified as forced if the *Wall Street Journal* reports that the CEO was fired, forced out, or left because of unspecified policy differences. Other departures are classified as forced if the CEO is under 60 and the announcement does not provide a normal voluntary reason or if the announcement reports retirement without at least six months of advance notice.
- **Voluntary turnover.** All other CEO turnovers are classified as voluntary after additional checks in business and trade press sources.
- **Sample sizes.** The classification yields 111 forced turnovers and 472 voluntary turnovers.

3.2 Matched control sample

The paper also constructs a matched control sample for the forced-turnover firms. For each forced-turnover firm, the authors select a NYSE firm with similar market capitalization and similar stock performance over the prior year, but with no CEO turnover in the prior three years. This produces 109 matched firms.

Interpretation. The matched control firms are important because they performed poorly, like forced-turnover firms, but their CEOs were not fired. This helps separate selling due merely to bad returns from selling associated with management being blamed for bad performance.

3.3 Institutional ownership data

Institutional ownership comes from the CDA/Spectrum database of 13F institutional investors. Institutions with at least $100 million in equity holdings must report their holdings to the SEC.

For each firm-quarter, the authors compute:

- the number of institutional investors holding the stock;
- the fraction of shares held by institutional investors;

- the number of institutional investors holding at least 1% of shares;
- institutional ownership by type of institution.

The five institution types are bank trust departments, insurance companies, investment companies, independent investment advisors, and other institutions.

3.4 Returns, accounting performance, and governance variables

- **Stock returns.** Quarterly returns are computed around the turnover announcement using CRSP and are market-adjusted using the CRSP equal-weighted NYSE index.
- **Accounting performance.** The accounting measure is EBIT divided by assets, industry-adjusted using the median firm in the same two-digit SIC industry.
- **Dividend cuts.** Dividend cuts and dividend eliminations are identified over the year before turnover.
- **Governance variables.** The paper measures CEO age, CEO share ownership, whether the CEO is a member of the founding family, and board composition. Directors are classified as inside, gray, or outside directors.

4 Methodology and empirical specifications

4.1 Raw changes in institutional ownership

For each event-time window around the CEO turnover quarter, the paper computes:

$$\Delta N_{it} = N_{i,t_2} - N_{i,t_1},$$

where N is the number of institutional investors, and

$$\Delta IO_{it} = IO_{i,t_2} - IO_{i,t_1},$$

where IO is the percentage of shares held by institutions.

The event quarter $t = 0$ is the quarter in which the CEO turnover is announced in the *Wall Street Journal*. The core pre-turnover window is the year before turnover, quarters -3 through 0 or -4 through -1, depending on the specification.

4.2 Abnormal changes in institutional ownership

To adjust for the general rise in institutional ownership and the fact that institutional ownership changes are positively related to returns, the paper estimates quarterly cross-sectional regressions for NYSE firms:

$$\Delta IO_{i,t} = a_t + b_t R_{i,t} + e_{i,t}.$$

The residual $e_{i,t}$ is interpreted as the abnormal change in institutional ownership. The paper computes abnormal changes for both the number of institutional investors and the fraction of shares owned by institutions.

Interpretation. If forced-turnover firms still show large negative residuals, the sell-off is not simply a mechanical result of poor stock returns or economywide institutional-ownership trends.

4.3 Testing explanations for institutional selling

The paper evaluates four hypotheses.

- **H1: Momentum trading.** Institutions may sell because returns are poor before forced turnover. The matched control sample and abnormal ownership regressions test whether poor returns alone explain the selling.
- **H2: Prudence constraints.** Institutions that prefer prudent securities may sell when firms cut dividends or become more volatile. The paper studies dividend cuts, return volatility, and institutional types such as bank trust departments.
- **H3: Informed trading.** Better-informed institutions may sell before other investors recognize the firm's prospects. The paper studies large institutional holders, independent investment advisors, and long-run returns following institutional buying and selling.
- **H4: Governance-cost concerns.** Some institutions may sell when direct intervention appears too costly. The paper studies founding-family CEOs, outside directors, and CEO ownership.

4.4 Board response regressions

The paper uses multinomial logit models to test whether changes in institutional ownership predict turnover outcomes. One model classifies observations as forced turnover, voluntary turnover, or no turnover. Another model separates turnover by whether the successor is an insider or outsider.

The key independent variables are the four institutional-ownership-change measures:

- change in number of institutional owners;
- change in percentage institutional ownership;
- abnormal change in number of institutional owners;
- abnormal change in percentage institutional ownership.

Controls include CEO age, founding-family status, dividend cuts, outside-director share, accounting performance, and stock performance.

5 Identification and interpretation

5.1 What the paper can identify

The paper identifies whether institutional ownership declines before forced CEO turnover and whether those changes are related to later board decisions. The evidence is strongest as an event-based empirical pattern with matched-sample and abnormal-change benchmarks.

5.2 Why forced CEO turnover is useful

Forced CEO turnover is useful because it marks a point at which the board appears to have lost confidence in the CEO. Observing institutional ownership before this point allows the paper to study whether investors had already reduced exposure.

5.3 Why the matched control sample matters

The matched control sample helps distinguish two stories. Under the simple momentum story, forced-turnover firms and similarly poor-performing matched firms should have similar institutional selling. In the paper, forced-turnover firms experience more selling, suggesting that poor returns alone do not fully explain the ownership shift.

5.4 Causality caveat

The paper does not prove that institutional selling mechanically causes boards to fire CEOs. Institutional selling could influence boards directly, could signal information to boards, or could be correlated with other unobserved factors such as media attention or private communication between investors and directors.

5.5 Interpretation of the ownership shift

The main outcome is not that all institutions sell. Instead, some institutions sell and others buy. The selling institutions tend to look more prudent, more informed, or more momentum-oriented, while the buying side includes individual investors and institutions with different preferences or information sets.

6 Tables and figures

6.1 Table 1: Institutional ownership two years before CEO turnover

Read:

- Table 1 compares institutional ownership for forced-turnover firms, matched control firms, and voluntary-turnover firms two years before the turnover.
- Forced-turnover firms have institutional percentage ownership similar to the matched sample, but fewer institutional investors than voluntary-turnover firms.
- Independent investment advisors and bank trust departments are the largest institutional owners.
- The matched control sample looks similar to the forced-turnover sample before the event, supporting its use as a benchmark.

Economic message: The forced-turnover firms do not begin with an unusually high institutional ownership percentage relative to the matched controls. The later change in ownership is therefore an event-window phenomenon, not simply a level difference.

Table 1
Institutional ownership for forced, matched control, and voluntary sample firms two years prior to the CEO turnover
This table reports the mean and median number of institutions holding shares in the corporation, percentage holdings by institutions, number of institution: holding at least 1% of the shares, and market capitalization for firms that experience a forced CEO turnover, a size- and return-matched control sample, anc firms that experience a voluntary CEO turnover. The ownership statistics are computed for forced and voluntary CEO turnovers at a point two years befor the CEO turnover (i.e., eight quarters before the quarter in which the turnover occurred) and for the matched control firm sample at the same time as thei: respective forced CEO turnover firm. The number of firms reported for each sample is the number of firms for which institutional ownership data are reportec in the CDA/Spectrum database two years before the turnover. Panel B reports institutional holdings by type of institution. The seventh through tenth column: report t-statistics for tests of differences in means and z-statistics from Wilcoxon rank-sum tests of whether the forced and matched control samples (seventl and eighth columns) or the forced and voluntary turnover samples (ninth and tenth columns) are drawn from the same distribution.

	Forced turnovers		Matched sample		Voluntary turnover		H_0: Force = Match		H_0: Force = Voluntary	
	Mean	Median	Mean	Median	Mean	Median	t-statistic	z-statistic	t-statistic	z-statistic
Panel A: All institutions										
No. of institutions	136	93	127	88	180	146	0.53	0.62	−3.04***	−4.09***
Ownership (%)	43.30	45.00	41.40	44.90	45.00	46.80	0.68	0.64	−0.83	−0.66
No. of institutions holding at least 1% of the shares	10.24	10.00	9.88	9.00	10.07	10.00	0.47	0.49	−0.29	0.11
Market cap. ($000)	2,121,590	724,278	1,995,378	639,888	4,156,708	1,759,091	0.22	0.18	−3.67***	−6.43***
Number of firms	102		101		455					
Panel B: By type of institution										
No. of institutions:										
Bank trust dept.	57	39	51	35	81	66	0.83	0.89	−4.21***	−4.59***
Insurance companies	12	9	11	9	16	13	0.46	0.78	−3.86***	−4.14***
Investment companies	7	4	7	5	10	8	−0.58	−0.69	−4.12***	−4.80***
Independent advisors	48	33	45	29	55	42	0.36	0.44	−1.26	−2.68***
Other institutional	13	9	13	9	18	16	0.45	0.51	−3.10***	−3.60***
Ownership (%):										
Bank trust dept.	12.4	11.1	11.4	11.4	15.0	14.7	0.96	0.54	−3.14***	−3.20***
Insurance companies	4.4	3.6	4.9	3.6	4.7	4.2	−0.80	−0.10	−0.90	−1.53
Investment companies	3.4	2.5	3.6	2.8	3.7	3.1	−0.48	−0.30	−0.88	−1.93*
Independent advisors	19.0	17.7	17.6	16.2	16.3	14.9	0.89	0.98	2.25**	2.19**
Other institutional	4.2	3.2	4.0	3.2	5.2	4.8	0.42	0.57	−2.35**	−2.58***

Figure 1: Table 1: Institutional ownership for forced, matched-control, and voluntary-turnover firms.

6.2 Table 2: Raw returns and institutional-ownership changes

Read:

- Table 2 shows that forced-turnover firms have large negative market-adjusted returns in the two years before turnover.
- The number of institutional investors falls by about 9.5 institutions in the year before forced turnover, while it rises for voluntary-turnover firms.
- Percentage institutional ownership falls by 4.34 percentage points in the two years before forced turnover, with the entire decline concentrated in the year before turnover.
- After turnover, institutional ownership begins to rise, but the rise is not clearly abnormal relative to marketwide trends.

Economic message: Institutions "vote with their feet" before forced CEO turnover. The sell-off is concentrated just before the CEO is removed.

Table 2
Abnormal returns and changes in total institutional ownership in four years around CEO turnover
For various periods in the four years around CEO turnovers for the samples of forced turnover, their matched control firms, and voluntary turnovers, thi table shows the mean market-adjusted compound abnormal returns (Panel A), the mean change in the number of institutional investors (Panel B), and th mean change in percentage ownership by institutional investors (Panel C). The numbers of observations in the period for the forced/matching control voluntary samples are given in the first row. T-statistics provided in the fourth and fifth rows of each panel test the null hypotheses that the forced turnover an matched control firm samples or the forced and voluntary turnover samples have equal means. T-statistics reported in parentheses test the null hypothesis tha the mean value does not differ from zero. The turnover occurs in quarter $t = 0$.

	Quarters (inclusive)									
	Level at $t = 0$	−7 thru 0	−7 thru −4	−3 thru −2	−1 thru 0	1 thru 2	3 thru 4	5 thru 8	1 thru 8	−7 thru 8
No. of firms (forced/match/vol.)		102/101/455	103/101/458	111/107/467	110/108/467	105/108/468	101/109/466	92/108/451	87/107/450	80/100/43(
Panel A: Mean market-adjusted compounded abnormal return (%)										
Forced		−30.22	−10.77	−10.43	−15.65	−5.30	3.59	6.02	8.21	−20.41
		(−8.01)***	(−3.30)***	(−4.14)***	(−5.42)***	(−1.92)*	(0.79)	(1.13)	(1.45)	(−3.51)**
Matched		−31.02	−16.28	−9.84	−10.63	−1.30	−3.21	3.54	−3.63	−31.02
		(−9.03)***	(−5.66)***	(−5.28)***	(−5.94)***	(−0.55)	(−1.48)	(0.91)	(−0.84)	(−6.43)**
Voluntary		−1.67	−0.43	−1.60	0.23	0.04	0.61	2.02	2.33	0.53
		(−0.95)	(−0.34)	(−1.94)*	(0.26)	(0.04)	(0.69)	(1.82)*	(1.47)	(0.22)
t-statistic (H_0: forced = matched)		0.16	1.27	−0.19	−1.48	−1.10	1.35	0.38	1.66*	1.40
t-statistic (H_0: forced = voluntary)		−6.87***	−2.96***	−3.33***	−5.26***	−1.83*	0.65	0.74	1.00	−3.33***

(a) Table 2, part 1: Abnormal returns and raw ownership changes.

Panel B: Mean change in number of institutional investors										
Forced	125	−6.26	2.04	−2.90	−6.57	−1.50	2.23	14.02	18.56	14.21
	($n = 111$)	(−1.81)	(0.81)	(−2.04)**	(−3.71)***	(−0.78)	(1.08)	(5.49)***	(4.33)***	(2.60)**
Matched	126	3.01	3.01	0.17	0.18	2.56	2.10	7.56	12.33	15.62
	($n = 108$)	(0.95)	(1.22)	(0.14)	(0.15)	(2.00)**	(1.72)*	(3.71)***	(4.21)***	(3.20)**
Voluntary	197	20.62	11.87	4.89	4.22	4.43	4.35	9.50	18.43	39.36
	($n = 469$)	(10.61)***	(9.59)***	(6.47)***	(5.25)***	(5.43)***	(5.77)***	(7.77)***	(10.48)***	(13.42)**
t-statistic (H_0: forced = matched)		−1.98**	−0.28	−1.67*	−3.20***	−1.76*	0.05	1.98**	1.20	−0.19
$t-$statistic (H_0: forced = voluntary)		−6.77***	−3.51***	−4.83***	−5.54***	−2.84***	−0.97	1.60	0.03	−4.06***
Panel C: Mean change in percentage institutional ownership (%)										
Forced	38.12%	−4.34	0.77	−1.83	−3.34	−0.09	−0.26	1.88	3.73	1.14
	($n = 111$)	(−3.06)***	(0.83)	(−2.47)**	(−3.57)***	(−0.12)	(−0.36)	(1.91)*	(2.80)***	(0.55)
Matched	43.00%	1.77	0.35	0.64	0.78	1.18	0.00	2.04	3.54	5.61
	($n = 108$)	(1.38)	(0.35)	(1.55)	(1.82)*	(2.12)**	(0.01)	(2.74)***	(4.12)***	(3.55)**
Voluntary	47.47%	2.60	1.45	0.64	0.63	1.14	0.49	1.42	3.27	5.65
	($n = 469$)	(7.11)***	(5.69)***	(2.93)***	(2.72)***	(4.41)***	(1.80)*	(4.57)***	(7.60)***	(10.34)**
t-statistic (H_0: forced = matched)		−3.20***	0.30	−2.91***	−4.00***	−1.38	−0.28	−0.13	0.12	−1.72*
t-statistic (H_0: forced = voluntary)		−4.74***	−0.71	−3.20***	−4.12***	−1.60	−0.98	0.45	0.33	−2.10**

*Indicates statistical significance at the 10% level; ** at the 5% level; *** at the 1% level.

(b) Table 2, part 2: Continued ownership changes.

6.3 Table 3: Abnormal ownership changes

Read:

- Table 3 adjusts ownership changes for marketwide trends and contemporaneous returns.
- The cross-sectional regressions confirm that institutional ownership changes are positively related to contemporaneous stock returns.
- Even after this adjustment, forced-turnover firms experience abnormal declines in both the number of institutions and institutional ownership percentage before turnover.
- The abnormal decline is larger than the raw decline, because institutional ownership was generally rising over the sample period.

Economic message: Momentum trading and poor returns explain part of institutional selling, but they do not fully explain the shareholder-composition shift before forced CEO turnover.

Table 3
Abnormal changes in institutional ownership in four years around CEO turnover
For each quarter between 1980 and 1996 (inclusive), we compute the abnormal change in the number of institutional investors for each NYSE security as th residual in a cross-sectional regression of the change in the number of institutions holding shares in the security over the quarter on the return over the quarter Similarly, we compute the abnormal change in the fraction of shares held by institutional investors by replacing the dependent variable with the change in th fraction of shares held by institutional investors over the quarter. The results of these 68 cross-sectional regressions are summarized in Panel A (on averag each regression contains 1,469 securities). The t-statistics in Panel A are computed from time-series standard errors. Panels B and C report, for various period in the four years around CEO turnovers, for the samples of forced turnovers, their matched control firms, and voluntary turnovers, the mean abnormal chang in the number of institutional investors (Panel B) and abnormal change in the fraction of shares held by institutional investors (Panel C). Abnormal changes i institutional ownership over multiple quarters are computed as the sum of the residuals over the quarters. The numbers of observations in the period for th forced/matching control/voluntary samples are given in the first row of Panel B. T-statistics provided in the last two rows of Panels B and C test the nul hypothesis that the forced turnover and matched control firm samples or the forced and voluntary turnover samples have equal means. T-statistics reported i parentheses test the null hypothesis that the mean value does not differ from zero. The turnover occurs in quarter $t=0$.

Panel A: Regression summary—Δinstitutional ownership $_{i,t}=\alpha_t+\beta_t(\text{return}_{i,t})+\varepsilon_{i,t}$

Dependent variable	Average intercept (t-statistic)	Average slope coefficient (t-statistic)	Average R^2
Change in No. of institutional investors	0.9450 (6.03)***	12.0361 (24.24)***	7.33%
Change in % institutional ownership	0.0022 (5.09)***	0.0298 (10.70)***	1.81%

	Quarters (inclusive)								
	−7 thru 0	−7 thru −4	−3 thru −2	−1 thru 0	1 thru 2	3 thru 4	5 thru 8	1 thru 8	−7 thru 8
No. of firms (forced/match/vol.)	102/101/455	103/101/458	111/107/467	110/108/467	105/108/468	101/109/466	92/108/451	87/107/450	80/100/436
Panel B: Mean abnormal change in number of institutional investors									
Forced turnover	−12.96	−2.56	−4.21	−7.02	−3.55	−0.30	7.59	6.18	−6.30
	(−4.21)***	(−1.12)	(−3.37)***	(−4.47)***	(−1.98)**	(−0.16)	(3.31)***	(1.60)	(−1.31)
Matched sample	−3.48	−0.96	−1.25	−0.93	−0.22	−0.44	1.46	0.88	−2.47
	(−1.27)	(−0.43)	(−1.20)	(−0.94)	(−0.18)	(−0.41)	(0.77)	(0.33)	(−0.57)

(a) Table 3, part 1: Abnormal ownership-change construction and early panels.

Voluntary turnover	8.79	5.91	2.10	1.11	1.65	1.43	3.79	7.05	16.02
	(4.91)***	(5.21)**	(3.04)**	(1.51)	(2.20)**	(2.12)**	(3.36)***	(4.41)***	(5.93)***
t-statistic (H_0: forced = matched)	−2.30**	−0.50	−1.82*	−3.29***	−1.54	0.07	2.05**	1.13	−0.59
t-statistic (H_0: forced = voluntary)	−6.11***	−3.33***	−4.42***	−4.68***	−2.68***	−0.86	1.48	−0.21	−4.03***
Panel C: Mean abnormal change in percentage institutional ownership (%)									
Forced turnover	−6.06	−0.50	−2.13	−3.39	−0.61	−0.89	0.45	0.70	−4.02
	(−4.53)***	(−0.56)	(−3.12)***	(−3.83)***	(−0.87)	(−1.31)	(0.48)	(0.58)	(−2.08)**
Matched sample	0.09	−0.67	0.28	0.52	0.38	−0.63	0.69	0.74	1.05
	(0.08)	(−0.70)	(0.69)	(1.29)	(0.71)	(−1.11)	(0.96)	(0.87)	(0.71)
Voluntary turnover	−0.19	0.06	−0.01	−0.15	0.41	−0.16	0.23	0.71	0.29
	(−0.54)	(0.24)	(−0.04)	(−0.65)	(1.69)*	(−0.61)	(0.75)	(1.74)*	(0.55)
t-statistic (H_0: forced = matched)	−3.44***	0.13	−3.04***	−4.02***	−1.12	−0.29	−0.20	−0.02	−2.08**
t-statistic (H_0: forced = voluntary)	−4.25***	−0.61	−2.97***	−3.55***	−1.38	−1.00	0.22	−0.01	−2.15**

*Indicates statistical significance at the 10% level; ** at the 5% level; *** at the 1% level.

(b) Table 3, part 2: Continued abnormal ownership changes.

6.4 Table 4: Dividend cuts, dividend eliminations, and ownership changes

Read:

- Table 4 partitions firms by whether they cut or eliminate dividends in the year before turnover.
- Forced-turnover firms are much more likely to cut or eliminate dividends than voluntary-turnover or matched-control firms.
- Institutions sell more when dividends are cut or eliminated, consistent with prudence constraints.
- However, forced-turnover firms still experience larger institutional selling than other firms even among firms that do not cut dividends.

Economic message: Dividend cuts explain part of the institutional exodus, but not all of it. Prudence concerns matter, yet the forced-turnover effect remains.

Table 4
Institutional ownership changes in the year preceding CEO turnover partitioned by dividend cuts and eliminations
This table shows the mean market-adjusted stock return and change in institutional ownership during the year preceding turnover for the forced, matchec control, and voluntary turnover samples partitioned according to whether dividends are cut or eliminated during the year preceding turnover. The statistics ar for samples of 110 forced turnovers, 107 matched control firms, and 465 voluntary turnovers. Twenty-eight of the forced turnover firms (10 dividend cuts, 1 dividend eliminations), seven of the matched control firms (5 dividend cuts, 2 dividend eliminations), and 29 of the voluntary turnover firms (20 dividend cuts 9 dividend eliminations) cut or eliminated their dividends in the year preceding turnover. The first column reports the averages for those firms in each sampl that did not cut or eliminate dividends. The second column reports the averages for those firms that reduced or eliminated their dividends. The third colum reports the averages for those firms that eliminated their dividends. *T*-statistics for tests that the average values equal zero are provided in parentheses. *T* statistics provided in the fourth and fifth row of each panel test the null hypothesis that the forced turnover and matched control firm samples or the forced an voluntary turnover samples have equal means. *T*-statistics provided in the last two columns test the null hypothesis that firms with and without dividend cut (i.e., first and second columns) or firms without dividend cuts and firms that eliminate their dividends (i.e., first and third columns) exhibit equal means.

	No dividend cut or elimination (n = 82/100/436)	Dividend cut or elimination (n = 28/7/29)	Dividend elimination (n = 18/2/9)	t-statistic (H_0: No cut = cut or elimination)	t-statistic (H_0: No cut = elimination)
Panel A: Average market-adjusted return					
Forced turnover	−13.73	−48.73	−56.70	4.74***	5.07***
	(−3.90)***	(−7.52)***	(−7.36)***		
Matched sample	−18.52	−22.80	−32.04	0.33	0.54
	(−7.59)***	(−1.80)	(−1.29)		
Voluntary turnover	0.05	−20.98	−25.15	4.69***	4.11***
	(0.04)	(−4.89)***	(−4.19)***		
t-statistic (H_0: forced = matched)	−1.12	1.82	0.95		
t-statistic (H_0: forced = voluntary)	3.68***	3.57***	3.23***		
Panel B: Average change in number of institutional owners					
Forced turnover	−3.05	−28.04	−25.83	3.99***	3.22***
	(−1.25)	(−4.86)***	(−3.88)***		
Matched sample	1.79	−3.84	−10.00	1.28	1.32
	(2.76)***	(−0.88)	(−1.25)		
Voluntary turnover	10.77	−15.66	−11.67	7.31***	7.14***
	(8.69)***	(−4.61)***	(−4.04)***		

(a) Table 4, part 1: Dividend cuts and ownership changes.

t-statistic (H_0: forced = matched)	3.22***	1.60	1.52		
t-statistic (H_0: forced = voluntary)	5.08***	1.85	1.95*		
Panel C: Average change in percentage total institutional ownership (%)					
Forced turnover	−2.80	−12.13	−15.93	3.02***	3.28***
	(−2.20)**	(−4.30)***	(−4.20)***		
Matched sample	1.79	−3.84	1.32	1.28	0.05
	(2.76)***	(−0.88)	(0.15)		
Voluntary turnover	1.16	2.66	6.14	−0.99	−2.32**
	(3.83)***	(1.80)*	(2.89)***		
t-statistic (H_0: forced = matched)	3.22***	1.60	1.80		
t-statistic (H_0: forced = voluntary)	3.03***	4.64**	5.08***		
Panel D: Average abnormal change in number of institutional owners					
Forced turnover	−6.89	−23.70	−19.39	3.05***	2.06*
	(−3.20)***	(−4.68)***	(−3.42)***		
Matched sample	−1.75	−7.97	−13.13	0.72	2.57
	(−1.07)	(−0.95)	(−3.19)***		
Voluntary turnover	4.68	−18.87	−14.40	6.81***	5.76***
	(4.09)***	(−5.79)***	(−4.63)***		
t-statistic (H_0: forced = matched)	1.90*	1.60	0.90		
t-statistic (H_0: forced = voluntary)	4.74***	0.80	0.77		
Panel E: Average abnormal change in percent institutional ownership (%)					
Forced turnover	−3.72	−10.79	−13.88	2.60**	2.84***
	(−3.08)***	(−4.42)***	(−4.12)***		
Matched sample	1.17	−4.53	−1.43	1.51	0.29
	(1.92)*	(−1.22)	(−0.16)*		
Voluntary turnover	−0.29	1.46	4.52	−1.31	−2.51**
	(−0.99)	(1.12)	(2.39)**		
t-statistic (H_0: forced = matched)	3.62***	1.41	1.30		
t-statistic (H_0: forced = voluntary)	2.77***	4.42***	4.76***		

*Indicates statistical significance at the 10% level; ** at the 5% level; *** at the 1% level.

(b) Table 4, part 2: Continued dividend-cut partitions.

6.5 Table 5: Return volatility around CEO turnover

Read:

- Table 5 reports daily return volatility by quarter around turnover.
- Forced-turnover firms have higher volatility than voluntary-turnover firms in every quarter shown.
- Volatility rises as forced turnover approaches and peaks around the turnover quarter.
- The comparison with matched-control firms is weaker, so volatility does not fully explain the ownership change.

Economic message: Risk and prudence concerns help explain why some institutions leave, but the evidence is not sufficient to make volatility the whole story.

6.6 Table 6: Institutional ownership changes by investor type

Read:

- Table 6 separates ownership changes by institution type.
- Bank trust departments are persistent sellers and account for a large share of the pre-turnover decline in institutional ownership.
- Independent investment advisors are major sellers in the year before turnover and account for a large share of the drop in institutional ownership percentage.
- Other investor categories do not show the same pattern of pre-turnover selling.

Economic message: The identity of the institution matters. The patterns fit both the prudence story for bank trust departments and the informed-trading story for independent investment advisors.

Table 5
Comparison of return volatilities in four years around CEO turnover
For each quarter in the four years around CEO turnover for the samples of forced turnovers, their matched control firms, and voluntary turnovers, this table shows the cross-sectional average standard deviation of daily returns (in percent). The numbers of observations in the period for the forced/matching/voluntary samples for each quarter are given in parentheses in the first column. *T*-statistics provided in the fourth and fifth columns test the null hypothesis that the forced turnover and matched firm samples or the forced and voluntary turnover samples have equal means, respectively. The turnover occurs in quarter $t=0$.

Quarter (*n*/*n*/*n*)	Forced turnovers	Matched sample	Voluntary turnovers	*t*-statistic H_0: force = match	*t*-statistic H_0: force = vol.
$t=-7$ (103/101/458)	2.09	2.33	1.78	−1.65	2.88***
$t=-6$ (103/101/458)	2.08	2.27	1.78	−1.28	3.06***
$t=-5$ (103/101/458)	2.19	2.31	1.80	−0.69	3.47***
$t=-4$ (103/101/458)	2.21	2.28	1.85	−0.45	2.98***
$t=-3$ (111/107/467)	2.37	2.17	1.90	1.34	3.80***
$t=-2$ (111/107/467)	2.56	2.37	1.94	0.90	4.24***
$t=-1$ (110/108/467)	2.89	2.47	1.90	1.91*	5.72***
$t=0$ (110/108/467)	3.28	2.48	1.90	3.15***	6.24***
$t=1$ (105/108/468)	2.98	2.60	1.87	1.41	4.84***
$t=2$ (105/108/468)	3.41	2.69	1.89	2.22**	5.48***
$t=3$ (101/109/466)	3.19	2.54	1.91	1.77*	3.88***
$t=4$ (101/109/466)	3.04	2.49	1.85	1.55	3.86***
$t=5$ (92/108/451)	2.58	2.36	1.74	0.94	4.37***
$t=6$ (92/108/451)	2.59	2.49	1.78	0.38	3.81***
$t=7$ (92/108/451)	2.60	2.63	1.75	−0.08	3.75***
$t=8$ (92/108/451)	2.60	2.43	1.75	0.63	3.82***

*Indicates statistical significance at the 10% level; ** at the 5% level; *** at the 1% level.

(a) Table 5: Return volatility around CEO turnover.

Table 6
Changes in institutional investor holdings by investor type in four years around forced CEO turnovers
For various periods in the four years around forced CEO turnovers, this table shows the mean change in the number of institutional investors (Panel A) and the mea change in institutional investors' percentage ownership (Panel B) by institutional investor type. The first column reports the level of institutional ownership (by class) a the time of turnover (quarter $t=0$). *T*-statistics reported in parentheses test the null hypothesis that the mean value does not differ from zero.

Level at $t=0$	Quarters (inclusive)								
	−7 thru 0	−7 thru −4	−3 thru −2	−1 thru 0	1 thru 2	3 thru 4	5 thru 8	1 thru 8	−7 thru 8
Panel A: Mean change in number of institutional investors									
	Bank trust departments								
49.47	−5.19	−0.80	−1.91	−2.91	−2.81	−0.45	1.48	−0.63	−4.91
	(−3.94)***	(−0.80)	(−3.52)***	(−4.47)***	(−3.41)***	(−0.60)	(1.88)*	(−0.41)	(−2.29)**
	Insurance companies								
10.52	−0.65	−0.50	0.18	−0.39	0.00	0.50	1.36	2.17	1.76
	(−1.81)	(−1.60)	(0.81)	(−1.79)	(0.00)	(1.85)*	(4.39)***	(5.48)***	(3.43)***
	Investment companies								
6.01	−0.35	0.09	−0.07	−0.40*	−0.02	0.62	1.62	2.48	2.48
	(−1.19)	(0.35)	(−0.34)	(−2.16)*	(−0.08)	(2.86)***	(4.19)***	(5.15)***	(4.51)***
	Independent investment advisors								
46.01	−0.14	2.52	−0.91	−2.34	1.08	1.25	8.84	13.32	13.83
	(−0.08)	(2.15)**	(−1.26)	(−2.69)***	(1.11)	(1.18)	(5.99)***	(5.89)***	(4.85)***
	Other institutional investors								
12.97	0.06	0.72	−0.20	−0.54	0.11	0.15	0.62	0.99	1.08
	(0.12)	(1.91)*	(−0.87)	(−2.14)*	(0.54)	(0.51)	(2.08)**	(2.21)**	(1.51)
Panel B: Mean change in percentage institutional investor ownership (%)									
	Bank trust departments								
10.17%	−1.95	−0.30	−0.55	−1.07	−0.54	−0.37	0.29	−0.28	−1.83
	(−3.97)***	(−0.83)	(−2.24)**	(−3.66)***	(−2.22)**	(−1.71)*	(0.94)	(−0.63)	(−2.50)**
	Insurance companies								
3.34%	−0.87	−0.66	−0.04	−0.16	0.10	−0.05	0.36	0.67	−0.23
	(−2.57)**	(−2.33)**	(−0.30)	(−0.81)	(0.54)	(−0.35)	(1.04)	(1.65)	(−0.39)
	Investment companies								
2.84%	−0.65	−0.33	0.09	−0.28	0.02	0.62	0.70	1.48	1.19
	(−2.17)**	(−1.32)	(0.48)	(−1.35)	(0.07)	(3.06)***	(1.59)	(3.79)***	(2.44)**
	Independent investment advisors								
17.45%	−1.13	1.71	−1.14	−1.84	0.37	−0.13	0.69	2.29	1.73
	(−1.14)	(2.70)***	(−2.07)**	(−3.04)***	(0.74)	(−0.30)	(0.93)	(2.56)**	(1.28)
	Other institutional investors								
4.32%	0.27	0.35	−0.19	0.01	−0.03	−0.32	−0.15	−0.43	0.27
	(0.78)	(1.61)	(−1.17)	(0.05)	(−0.19)	(−1.47)	(−1.02)	(−2.07)*	(0.65)

*Indicates statistical significance at the 10% level; ** at the 5% level; *** at the 1% level.

(b) Table 6: Ownership changes by investor type.

6.7 Table 7: Institutional ownership changes by size of holding

Read:

- Table 7 partitions institutions by whether they held at least 1% of shares two years before forced turnover.
- Institutions with larger holdings account for more than half of institutional ownership at the forced-turnover firms.
- These large-holding institutions reduce ownership much more than institutions with smaller holdings.
- Because large holders have stronger incentives to acquire information, the pattern supports the informed-trading interpretation.

Economic message: The selling is concentrated among institutions most likely to be informed about firm prospects.

6.8 Table 8: Long-run returns after institutional buying and selling

Read:

- Table 8 forms quarterly NYSE portfolios based on changes in institutional ownership.
- Stocks most heavily purchased by institutions outperform stocks most heavily sold by institutions over subsequent horizons of up to four years.
- The result holds for changes in the number of institutions, changes in percentage ownership, and abnormal versions of both measures.
- This supports the idea that institutional trades contain information.

Economic message: Institutional investors appear, on average, to trade in the right direction. This makes their selling before forced turnover more informative.

Table 7
Changes in institutional ownership over two years preceding forced CEO turnover, sorted by size of institutions' holdings
Institutions are partitioned based on size of holdings in the firm, including those with holdings of less than 1% and those with holdings of 1% or more of th outstanding shares two years prior to forced CEO turnover. The first row reports the mean cross-sectional fraction of shares held by institutions within eacl group two years prior to CEO turnover. The second row reports the mean cross-sectional percentage change in institutional ownership attributed to eacl group. The third through fifth rows report, respectively, the fraction of institutions within each group exhibiting, in the two years prior to turnover, decrease in their positions, increases in their positions, and no change in their positions. The fourth column reports t-statistics for tests that the values in the first twc columns are equal. The first t-statistic is a paired t-statistic and the last three are for tests of differences in means. The statistics in the first two rows ar computed at the firm level for the 102 firms with forced turnover. The statistics in rows three through five are computed at the institution-firm level with th number of institution-firm observations reported in the last row. The turnover occurs in quarter $t=0$.

	Institutions with $\geqslant$1% ownership	Institutions with < 1% ownership	Institutions with no ownership at $t=-8$	t-statistic for test that the values in th first and second columns are equal
Institutional ownership at $t=-8$ ($n=102$)	26.13%	17.16%		
Change in ownership in quarters $t=-7$ to $t=0$ ($n=102$)	−13.50%	−3.48%	12.64%	−8.34***
Institutions with negative change	80.36%	74.41%	0.00%	−4.62***
Institutions with positive change	18.77%	25.04%	100.00%	4.94***
Institutions with no change	0.86%	0.55%	0.00%	−1.05
Number of observations	1,044	12,850	5,210	

*Indicates statistical significance at the 10% level; ** at the 5% level; *** at the 1% level.

(a) Table 7: Ownership changes by holding size.

Table 8
Long-term performance and changes in institutional ownership
This table reports the results of an analysis of the relation between institutional ownership changes and long-term performance. Each quarter, NYSE stock: are partitioned into change in institutional ownership quintiles. A market-adjusted return is calculated for each stock-quarter as the quarterly return on th stock less the CRSP equal-weighted NYSE return. Market-adjusted returns for each stock are then compounded up to four years. Cross-sectional averages ar then computed for firms in each institutional ownership change portfolio. The table reports the time-series average of these cross-sectional averages for th extreme quintiles, including the quintile with the largest increase in institutional ownership and the quintile with the largest decrease. The third row of eacl panel is the mean difference between the first two rows. T-statistics (computed from time-series standard errors) for tests that the average values equal zero ar provided in parentheses

	Quarter +1	One year forward return ($n=68$)	Two years forward return ($n=68$)	Three years forward return ($n=68$)	Four years forwarc return ($n=64$)
Panel A: Quintiles formed on changes in number of institutional investors					
Largest increase in number of institutions	0.34	1.29	1.72	2.94	5.42
	(0.85)	(2.06)**	(1.70)	(2.39)**	(3.83)***
Largest decrease in number of institutions	−0.38	−2.12	−1.41	−0.61	0.68
	(−1.42)	(−4.15)***	(−1.94)*	(−0.67)	(0.60)
Difference	0.72	3.41	3.13	3.55	4.75
	(1.24)	(3.72)***	(2.32)**	(2.29)**	(2.64)**
Panel B: Quintiles formed on changes in fraction of shares held by institutional investors					
Largest increase in % institutional ownership	0.06	0.62	0.72	1.43	2.40
	(0.26)	(1.51)	(1.32)	(2.06)**	(2.84)***
Largest decrease in % institutional ownership	−0.24	−1.54	−1.24	−0.85	−0.89
	(−0.71)	(−3.25)***	(−2.06)**	(−1.21)	(−0.89)
Difference	0.30	2.17	1.96	2.28	3.29
	(0.73)	(3.62)***	(2.58)**	(2.34)**	(2.56)**
Panel C: Quintiles formed on abnormal changes in number of institutional investors					
Largest increase in abnormal number of institutions	0.45	1.04	1.74	2.91	5.21
	(1.39)	(1.80)*	(2.03)**	(2.89)***	(4.45)***
Largest decrease in abnormal number of institutions	−0.32	−0.73	−1.32	1.13	2.41
	(−1.42)	(−1.68)	(−0.21)	(1.32)	(2.36)**
Difference	0.77	1.77	1.87	1.77	2.79
	(1.76)*	(2.43)**	(1.76)*	(1.36)	(1.94)*
Panel D: Quintiles formed on abnormal changes in fraction of shares held by institutional investors					
Largest increase in abnormal change in % institutional	0.24	0.50	0.73	1.56	2.36
	(0.84)	(1.09)	(1.27)	(2.02)**	(2.59)**
Largest decrease in abnormal change in % institutional	−0.11	−0.93	−0.74	−0.40	−0.01
	(−0.47)	(−2.26)**	(−1.27)	(−0.60)	(−0.08)
Difference	0.36	1.44	1.47	1.97	2.43
	(1.22)	(2.53)**	(2.05)**	(2.14)**	(2.18)**

*Indicates statistical significance at the 10% level; ** at the 5% level; *** at the 1% level.

(b) Table 8: Long-run returns after institutional trading.

6.9 Table 9: Governance characteristics and institutional selling

Read:

- Table 9 regresses ownership changes on governance variables and controls.
- Institutional selling is somewhat larger when the CEO is a member of the founding family.
- Outside-director share and CEO ownership do not explain much of the ownership change.
- The forced-turnover sample still shows larger institutional selling even after controlling for governance, dividend cuts, accounting performance, stock returns, and size.

Economic message: There is limited support for the idea that institutions sell because direct governance intervention is too costly. The strongest governance-related evidence is for founding-family CEOs.

6.10 Table 10: Institutional ownership changes and forced turnover

Read:

- Table 10 estimates multinomial logit models distinguishing forced turnover, voluntary turnover, and matched no-turnover observations.
- Without ownership-change controls, older CEOs and founding-family CEOs are less likely to be forced out, and better stock or accounting performance reduces forced-turnover likelihood.
- When ownership-change variables are added, declines in institutional ownership significantly predict forced turnover.
- The ownership-change variables absorb much of the explanatory power of prior stock returns.

Economic message: Institutional selling is not merely an outcome around forced turnover; it helps predict which poorly performing CEOs are forced out.

Table 9
Regression of change in institutional ownership on corporate governance characteristics and other variables
This table reports results from OLS regressions of the change in institutional ownership in the year prior to turnover on firm characteristics. The sample includes 570 firms with adequate data that experience CEO turnover or are a matched control firm for forced CEO turnovers (65 forced, 398 voluntary, and 107 matching). The change in institutional ownership is represented by four variables used in the different regressions: change in the number of institutional owners, change in the fraction of shares held by institutional investors, abnormal change in the number of institutional owners, and abnormal change in the fraction of shares held by institutional investors. The firm characteristics include a matching firm dummy variable (one if the firm is in the matching firm control sample and zero otherwise), a voluntary turnover firm dummy variable (one if the firm is in the voluntary turnover sample and zero otherwise), founding family dummy (one if CEO is a member of the founding family and zero otherwise), the fraction of directors who are outsiders, the fraction of shares held by the CEO, a dummy variable indicating a dividend cut in the year preceding turnover, industry-adjusted level of accounting performance (EBIT/book assets) in the year preceding turnover, and the market-adjusted stock return in the year preceding turnover. The *t*-statistics are reported in parentheses.

Explanatory variable	Dependent variable			
	Δ no. of institutions	Δ % institutional ownership	Abnormal Δ no. of institutions	Abnormal Δ % institutional ownership
Constant	−25.45**	0.05	−21.87*	0.39
	(−2.02)	(1.14)	(−1.77)	(1.03)
Matching sample dummy	10.11**	0.06***	10.33**	0.06***
	(2.85)	(5.14)	(2.99)	(5.19)
Voluntary turnover dummy	7.91**	0.05***	7.32**	0.04***
	(2.67)	(5.29)	(2.53)	(4.97)
Founding family dummy	−6.24*	−0.02	−6.04*	−0.01
	(−1.89)	(−1.51)	(−1.88)	(−1.39)
% outside directors	−2.06	−0.003	−2.00	0.00
	(−0.27)	(−0.13)	(−0.26)	(0.16)
% shares owned by the CEO	11.29	0.03	12.18	0.04
	(0.59)	(0.43)	(0.65)	(0.66)
Dividend cut dummy	−13.71***	0.01	−13.02***	0.01
	(−3.94)	(0.87)	(−3.83)	(0.76)
Industry-adjusted EBIT/assets	−3.45	−0.01	−1.12	0.01
	(−0.50)	(−0.21)	(−0.17)	(0.31)
Market-adjusted return	46.10***	0.06***	34.94***	0.04**
	(12.80)	(5.53)	(9.93)	(3.39)
Log (market capitalization)	2.10**	−0.01**	1.45*	−0.59**
	(2.73)	(−2.24)	(1.94)	(−2.55)
N	570	570	570	570
R^2	0.35	0.11	0.26	0.08

*Indicates statistical significance at the 10% level; ** at the 5% level; *** at the 1% level.

(a) Table 9: Governance variables and institutional ownership changes.

Table 10
Multinomial logit regression of turnover on firm characteristics
This table shows results from multinomial logistic regression models with type of turnover (zero for forced, one for voluntary, and two for no turnover (i.e. matching control firm sample)) as the dependent variable and firm characteristics as the independent variables for a sample of 575 firms with adequate data (6 forced turnovers, 402 voluntary turnovers, and 107 no-turnover matching control firms). The firm characteristics include CEO age dummy (one if CEO age i greater than 59 and zero otherwise), founding family dummy (one if CEO is a member of the founding family and zero otherwise), a dummy variable indicatin a dividend cut in the year preceding turnover, the percentage of outside directors, industry-adjusted level of accounting performance (EBIT/book assets) in th year preceding turnover, the market-adjusted stock return in the year preceding turnover, and the change in institutional ownership in the four quarter preceding turnover. The change in institutional ownership is represented by four variables used in the different regressions: change in the number o institutional owners (Model 2), change in the fraction of shares held by institutional investors (Model 3), abnormal change the in number of institutiona owners (Model 4), and abnormal change in the fraction of shares held by institutional investors (Model 5). The *t*-statistics are reported in parentheses.

	Model									
	(1)		(2)		(3)		(4)		(5)	
Explanatory variable	Voluntary	Matching	Voluntary	Matching	Voluntary	Matching	Voluntary	Matching	Voluntary	Matching
Constant	0.901	0.054	0.985	0.107	1.186	0.273	1.091	0.231	1.260	0.398
	(0.92)	(0.05)	(0.97)	(0.10)	(1.18)	(0.25)	(1.08)	(0.21)	(1.25)	(0.36)
CEO age dummy	2.059***	−0.138	2.031***	−0.149	1.940***	−0.257	2.043***	−0.140	1.948***	−0.254
	(6.90)	(−0.40)	(6.66)	(−0.42)	(6.37)	(−0.72)	(6.69)	(−0.40)	(6.41)	(−0.72)
Founding family dummy	2.021*	3.158***	2.048*	3.172***	1.98*	3.100***	2.068**	3.179***	1.962*	3.068**
	(1.94)	(3.04)	(1.96)	(3.04)	(1.89)	(2.96)	(1.98)	(3.05)	(1.87)	(2.93)
Dividend cut dummy	−0.533	−1.333**	−0.345	−1.103*	−0.592	−1.361**	−0.333	−1.081*	−0.583	−1.367**
	(−1.19)	(−2.41)	(−0.76)	(−1.95)	(−1.30)	(−2.41)	(−0.73)	(−1.91)	(−1.28)	(−2.42)
% outside directors	−0.576	0.296	−0.848	0.040	−0.881	−0.050	−0.860	−0.005	−0.917	−0.084
	(−0.45)	(0.21)	(−0.64)	(0.03)	(−0.67)	(−0.03)	(−0.65)	(−0.00)	(−0.70)	(−0.06)

(b) Table 10, part 1: Multinomial logit models of turnover type.

6.11 Table 11: Turnover frequencies by institutional-ownership change

Read:

- Table 11 compares firms where institutional ownership rose with firms where institutional ownership fell in the year before turnover.
- Forced turnover is more frequent when institutional ownership declines.
- Outside succession is also more frequent when institutional ownership declines.
- Among forced turnovers, outsider replacement is much more common after institutional ownership falls.

Economic message: The decline in institutional ownership is associated not only with CEO firing, but also with a more drastic succession decision: choosing an outsider.

Industry-adjusted EBIT/assets	2.264*	−0.389	2.454*	−0.042	2.503*	−0.032	2.477*	0.015	2.487*	−0.050
	(1.74)	(−0.27)	(1.83)	(−0.03)	(1.89)	(−0.02)	(1.85)	(0.01)	(1.87)	(−0.03)
Market-adjusted return	1.400**	−0.746	0.463	−1.621*	0.964	−1.493*	0.641	−1.494*	1.102*	−1.310*
	(2.24)	(−1.05)	(0.63)	(−1.95)	(1.46)	(−1.95)	(0.93)	(−1.90)	(1.70)	(−1.74)
Δ number of institutions			0.018**	0.018**						
			(2.46)	(2.16)						
Δ % institutional ownership					4.354**	8.361**				
					(1.99)	(3.29)				
Abnormal Δ number of institutions							0.02***	0.02**		
							(2.60)	(2.40)		
Abnormal Δ % institutional ownership									3.649*	8.673***
									(1.65)	(3.33)
Number of forced turnovers	66.0		64.0		64.0		64.0		64.0	
Number of voluntary turnovers	402.0		399.0		399.0		400.0		399.0	
Number of matching firm observations	107.0		104.0		104.0		104.0		104.0	
Model χ^2	204.1***		200.7***		206.1***		202.0***		206.9***	

*Indicates statistical significance at the 10% level; ** at the 5% level; *** at the 1% level.

(a) Table 10, part 2: Continued multinomial logit results.

Table 11
CEO turnover frequencies by change in institutional ownership during the year preceding turnover
Turnover is classified as forced if the incumbent CEO departs prior to age 60 and does not leave for other employment or for health reasons or if the *Wall Street Journal* reports that the CEO was forced from the position. All other turnovers are classified as voluntary. An outside succession is a succession in which the new CEO has been employed at the firm for one year or less at the time of the turnover.

	Increase in institutional ownership			Decrease in institutional ownership			X^2 statistic for test that the percentages of total are equal for positive and negative institutional ownership changes
	N	Percent of total (%)	Percent of forced (%)	N	Percent of total (%)	Percent of forced (%)	
All turnovers	334			237			
Forced turnover	46	13.77		62	26.16		10.60***
Outside succession	52	15.57	41.30	59	24.89	66.13	5.73***
Forced/outside succession	19	5.69		41	17.30		16.70***

*Indicates statistical significance at the 10% level; ** at the 5% level; *** at the 1% level.

(b) Table 11: Turnover frequencies by ownership change.

6.12 Table 12: Ownership changes and outsider succession

Read:

- Table 12 estimates multinomial logit models that separate voluntary/inside, voluntary/outside, forced/inside, and forced/outside succession.
- The key result is that declines in institutional ownership are associated with a higher likelihood of forced/outside succession.
- The effect is strongest for the change in the number of institutional investors and the abnormal change in the number of institutional investors.
- Outside-director share also helps predict outside succession.

Economic message: Boards appear more likely to make a clean break with the old regime when institutions have been selling the stock.

Table 12
Regression of relation between turnover type, origin of successor, and changes in institutional ownership
This table shows results from multinomial logistic regression models with four turnover outcomes (voluntary/inside, voluntary/outside, forced/inside, anc forced/outside) as the dependent variables and firm characteristics as the independent variables. The firm characteristics include CEO age dummy (one if CEC age is greater than 59 and zero otherwise), a dummy variable indicating a dividend cut in the year preceding turnover, the percentage of outside directors industry-adjusted level of accounting performance (EBIT/book assets) in the year preceding turnover, the market adjusted stock return in the year precedin turnover, and the change in institutional ownership in the four quarters preceding turnover. The change in institutional ownership is represented by fou variables used in the different regressions: change in the number of institutional owners (Model 1), change in the fraction of shares held by institutiona investors (Model 2), abnormal change in the number of institutional owners (Model 3), and abnormal change in the fraction of shares held by institutiona investors (Model 4). The *t*-statistics are reported in parentheses.

	Model 1			Model 2		
	Turnover outcome			Turnover outcome		
Explanatory variable	Vol/out	Force/ins	Force/out	Vol/out	Force/ins	Force/ou
Constant	−5.01***	−2.31*	−3.78**	−5.10***	−2.40*	−4.22**
	(−3.70)	(−1.72)	(−2.44)	(−3.73)	(−1.79)	(−2.74)
CEO age dummy	−0.74*	−2.55***	−1.88***	−0.72*	−2.51***	−1.75**
	(−1.96)	(−5.89)	(−4.61)	(−1.91)	(−5.77)	(−4.33)
Dividend cut dummy	−0.28	−0.35	0.75	−0.35	−0.22	1.14**
	(−0.42)	(−0.47)	(1.40)	(−0.53)	(−0.30)	(2.13)
% outside directors	4.97***	1.98	3.70*	5.04***	2.05	3.99**
	(2.89)	(1.11)	(1.84)	(2.91)	(1.14)	(2.01)
Industry-adjusted EBIT/assets	−2.58*	−2.09	−2.64	−2.59*	−2.23	−2.79
	(−1.86)	(−1.21)	(−1.48)	(−1.84)	(−1.30)	(−1.61)
Market-adjusted return	−0.05	−1.54	0.28	−0.17	−1.34	−0.67
	(−0.06)	(−1.43)	(0.29)	(−0.25)	(−1.39)	(−0.75)
Δ number of institutions	0.00	−0.00	−0.02***			
	(0.08)	(−0.38)	(−2.81)			
Δ % institutional ownership				2.70	−4.28	−5.03*
				(0.96)	(−1.53)	(−1.82)
N (turnovers)		462			462	
N (voluntary/inside)		353			353	
N (voluntary/outside)		45			45	
N (forced/inside)		31			31	
N (forced/outside)		33			33	
Model χ^2		108.22***			106.75***	

(a) Table 12, part 1: Successor-origin multinomial logit models.

Table 12 (*continued*)

	Model 3			Model 4		
	Turnover outcome			Turnover outcome		
Explanatory variable	Vol/out	Force/ins	Force/out	Vol/out	Force/ins	Force/ou
Constant	−5.01***	−2.33*	−3.88**	−5.10***	−2.44*	−4.28***
	(−3.71)	(−1.74)	(−2.52)	(−3.72)	(−1.82)	(−2.79)
CEO age dummy	−0.74*	−2.55***	−1.88***	−0.73*	−2.51***	−1.75***
	(−1.95)	(−5.89)	(−4.61)	(−1.92)	(−5.79)	(−4.33)
Dividend cut dummy	−0.28	−0.35	0.75	−0.38	−0.24	1.12**
	(−0.42)	(−0.48)	(1.41)	(−0.57)	(−0.33)	(2.11)
% outside directors	4.98***	1.97	3.64*	5.09***	2.04	4.00**
	(2.90)	(1.11)	(1.82)	(2.92)	(1.14)	(2.01)
Industry-adjusted EBIT/assets	−2.57*	−2.09	−2.72	−2.61*	−2.22	−2.81
	(−1.86)	(−1.21)	(−1.52)	(−1.86)	(−1.29)	(−1.61)
Market-adjusted return	−0.06	−1.56	0.01	−0.14	−1.50	−0.83
	(−0.08)	(−1.53)	(0.01)	(−0.21)	(−1.59)	(−0.94)
Abnormal Δ number of institutions	0.00	−0.00	−0.03***			
	(0.14)	(−0.43)	(−2.85)			
Abnormal Δ % institutional ownership				3.93	−3.39	−4.24
				(1.36)	(−1.21)	(−1.52)
N (turnovers)		462			462	
N (voluntary/inside)		353			353	
N (voluntary/outside)		45			45	
N (forced/inside)		31			31	
N (forced/outside)		33			33	
Model χ^2		108.49***			106.33***	

*Indicates statistical significance at the 10% level; ** at the 5% level; *** at the 1% level.

(b) Table 12, part 2: Continued successor-origin logit models.

7 Short summary

- The paper studies whether institutional investors sell before forced CEO turnover and whether this selling is related to board decisions.

- The sample covers large NYSE firms with CEO turnovers from 1982 to 1993: 111 forced turnovers and 472 voluntary turnovers.
- The paper also constructs a matched control sample of poor-performing firms without CEO turnover.
- Institutional investors reduce their holdings before forced CEO turnover, especially in the year immediately before the CEO is fired.
- The sell-off is not universal: some institutions sell, while other institutions and individual investors buy.
- Poor returns and momentum trading explain part, but not all, of the selling.
- Dividend cuts, higher volatility, and selling by bank trust departments support a prudence-based explanation.
- Selling by large holders and independent investment advisors supports an informed-trading explanation.
- Governance-cost explanations receive weaker support, mainly through evidence involving founding-family CEOs.
- Institutional selling predicts forced CEO turnover and the appointment of an outside successor.

8 Conclusion

8.1 Core conclusion

Parrino, Sias, and Starks show that institutional investors often sell before forced CEO turnover. This supports the idea that institutions can express dissatisfaction through exit rather than activism. The ownership change is large, concentrated before the forced turnover event, and concentrated among particular types of institutions.

8.2 Economic intuition

Institutional investors differ in mandates, information, and incentives. Some institutions may need to avoid risky or imprudent securities. Others may be better informed and therefore sell earlier. Still others may sell because the stock has performed poorly. When these institutions leave, the firm's shareholder base changes in ways boards may notice.

8.3 Incremental contribution revisited

The paper's key contribution is to bring the "Wall Street Rule" into empirical corporate governance. It shows that the decision to sell can be an important governance mechanism and that the market for corporate control through trading can operate before formal board action.

8.4 Implications

The results imply that institutional trading can provide a signal to boards. A decline in institutional ownership may indicate that sophisticated shareholders have lost confidence in management. The board may respond by firing the CEO and by appointing an outsider to restore investor confidence.

8.5 Limitations

The evidence is not a clean randomized causal estimate. Institutional selling may influence boards, but it may also reflect private communication, public information, or an unobserved shock that affects both investors and directors. The paper is careful to interpret the board-decision evidence as consistent with influence rather than definitive proof of causation.

8.6 Takeaway

The enduring takeaway is that institutional investors govern not only by voice but also by exit. Around forced CEO turnover, exit changes the shareholder base, reveals information, and is associated with board decisions to replace the CEO and sometimes to choose an outside successor.